

Practice Problems

Use the integral test to determine whether the following series converge or diverge.

1. $\sum_{n=1}^{\infty} \frac{1}{\sqrt[5]{n}}$

6. $\sum_{n=1}^{\infty} \frac{5 - 2\sqrt{n}}{n^3}$

11. $\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$

2. $\sum_{n=1}^{\infty} \frac{1}{(2n+1)^3}$

7. $\sum_{n=1}^{\infty} \frac{n^2}{n^3+1}$

12. $\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$

3. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+4}}$

8. $\sum_{n=1}^{\infty} \frac{3n+2}{n(n+1)}$

13. $\sum_{n=3}^{\infty} \frac{n^2}{e^n}$

4. $\sum_{n=1}^{\infty} \frac{n+2}{n+1}$

9. $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^3}$

14. $\sum_{n=1}^{\infty} \frac{1}{n^3+n}$

5. $\sum_{n=1}^{\infty} n^{-1.4} + 3n^{-1.2}$

10. $\sum_{n=1}^{\infty} \frac{1}{n^2 - 4n + 5}$

15. $\sum_{n=1}^{\infty} \frac{n}{n^4+1}$

Answers to Practice Problems

1. Diverge

6. Converge

11. Converge

2. Converge

7. Diverge

12. Converge

3. Diverge

8. Diverge

13. Converge

4. Diverge

9. Converge

14. Converge

5. Converge

10. Converge

15. Converge